INVERSE FUNCTIONS QUIZ

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Name: Solutions

Solve the following equations for x.

1. $\log_4(x+3) + \log_4(x-3) = 2$

$$2 = \log_4(x+3) + \log_4(x-3) = \log_4((x+3)(x-3)) = \log_4(x^2-9)$$

$$\Rightarrow \quad 4^2 = 4^{\log_4(x^2-9)} = x^2 - 9$$

$$\Rightarrow \quad x^2 = 16 + 9 = 25$$

$$\Rightarrow \quad \boxed{x = 5} \quad (\log_4(x+3) \text{ and } \log_4(x-5) \text{ are not defined at } -5).$$

2. $e^x = \left(\dfrac{e^{\sqrt{x}}}{e^{\frac{1}{2}}}\right)^2$

$$e^x = \frac{e^{2\sqrt{x}}}{e^{2(\frac{1}{2})}} = e^{2\sqrt{x}-1}$$

$$\Rightarrow \ln(e^x) = \ln(2\sqrt{x}-1)$$

$$\Rightarrow \quad x = 2\sqrt{x} - 1$$

$$\Rightarrow \quad x - 2\sqrt{x} + 1 = (\sqrt{x}-1)^2 = 0$$

$$\Rightarrow \quad \sqrt{x} = 1$$

$$\Rightarrow \quad \boxed{x = 1}$$